

Evaluating CNN-LSTM and GNN-GRU Hybrid Models for Enhanced Air Quality Prediction and Dynamic Forecasting



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Introduction

- According to the World Bank, almost half of the global population inhabited urban areas in recent years, and this figure is expected to extend up to 70% by 2050 [1].
- Among the most hazardous air pollutants are fine particles with an aerodynamic diameter of 2.5 μm or less, known as PM2.5 [2].
- Machine learning techniques, including Artificial Neural Networks (ANN) and Support Vector Machines (SVM), have been applied to improve prediction accuracy [3, 4].
- In the present study, air quality and meteorological secondary data were used to predict pollution levels for future time steps by using various deep learning techniques.

Study Area and Data Description

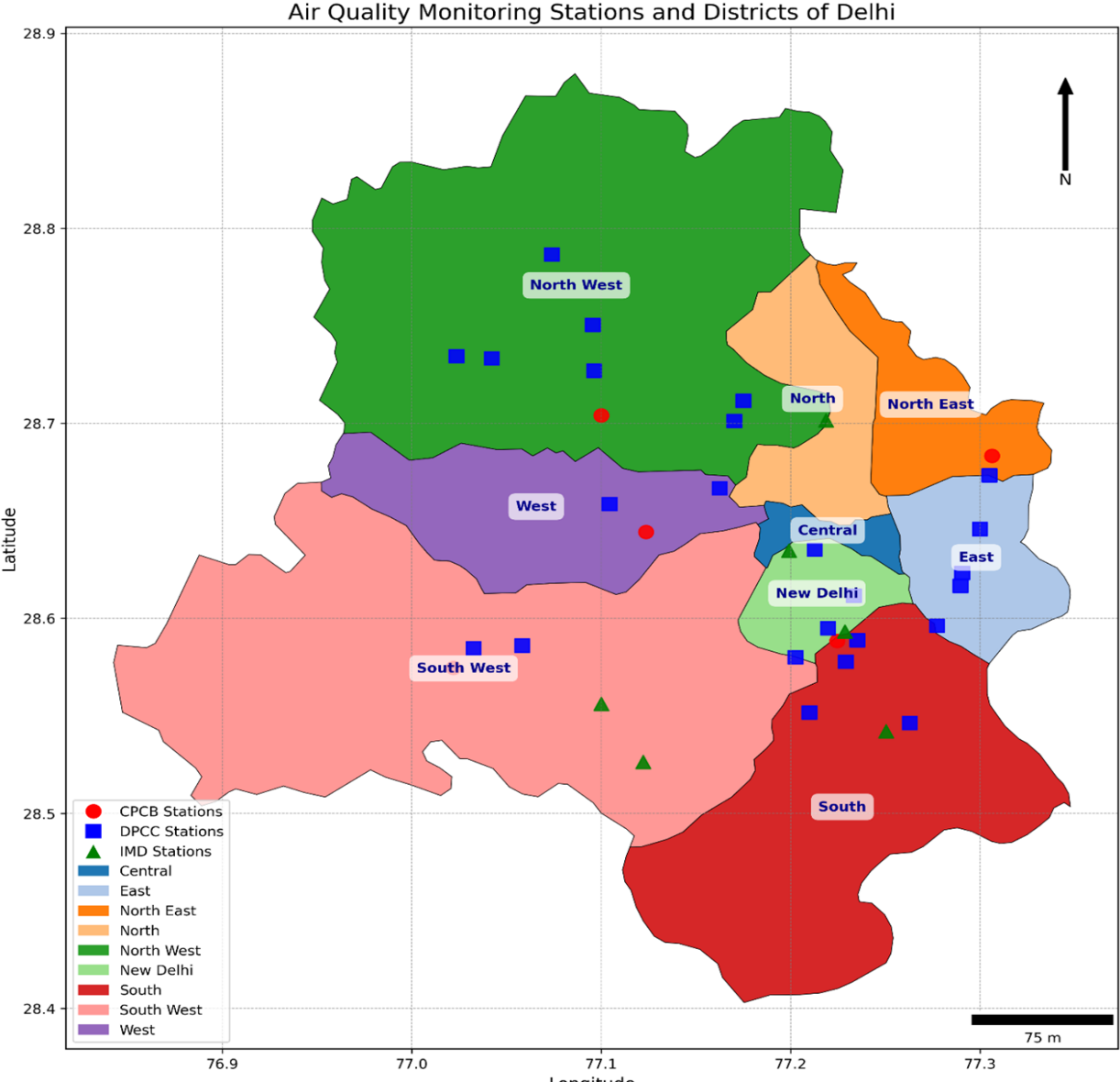


Figure 1: Map showing air quality monitoring stations in Delhi.

Feature	Description
Timestamp	Date and time of the observation
PM2.5	Particulate matter concentration
CLRSKY SFC SW DWN	Clear sky solar radiation
T2M	Temperature measured at 2 meters above the ground
PRECTOTCORR	Corrected total precipitation
PS	Surface-level atmospheric pressure
WS2M and WS10M	Wind speed at 2- and 10-meters height
WD2M and WD10M	Wind direction at 2- and 10-meters height
RH2M	Relative humidity at 2 meters above the ground

- 35 CAAQMS stations in Delhi are managed by CPCB and state bodies.
- Stations are located across residential, industrial, traffic, and green zones to capture spatial variations.
- Figure 1 shows the geographic distribution of these stations.
- Collected 5 years of hourly PM2.5 data from CPCB.
- Gathered meteorological data (temperature, relative humidity, wind speed, precipitation) from NASA POWER.
- Datasets help explore the interplay between air quality and meteorological conditions.

Methodology

- Before Modelling: Select key features for PM2.5 prediction and reduce data dimensionality.
- Step 1: *Data Preparation*: PM2.5 + meteorological variables (T2M, RH2M, WS2M); cleaned and pre-processed.
- Step 2: *Optimized Features*:
 - Rolling stats: 6-hour mean & std of PM2.5
 - Interaction term: T2M $\times$ RH2M
- Step 3: *PCA*: Top 3 components captured 81.7% variance.
- Step 4: Feature Importance: *Key Findings*:
 - T2M $\times$ RH2M interaction was most important.
 - Meteorological features (T2M, RH2M, WS2M) showed low importance.
 - Rolling stats had negligible contribution to PM2.5 prediction.

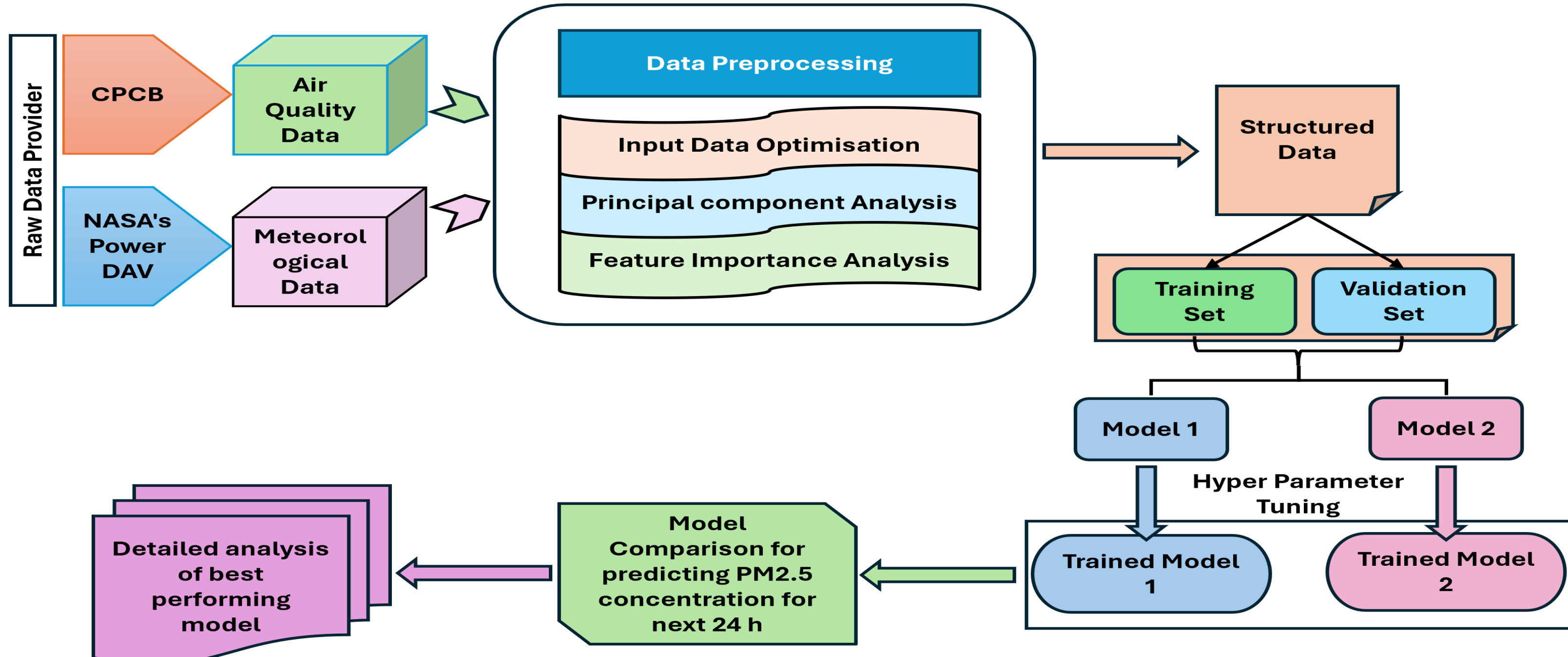


Figure 2: Methodology for data processing using a deep learning approach.

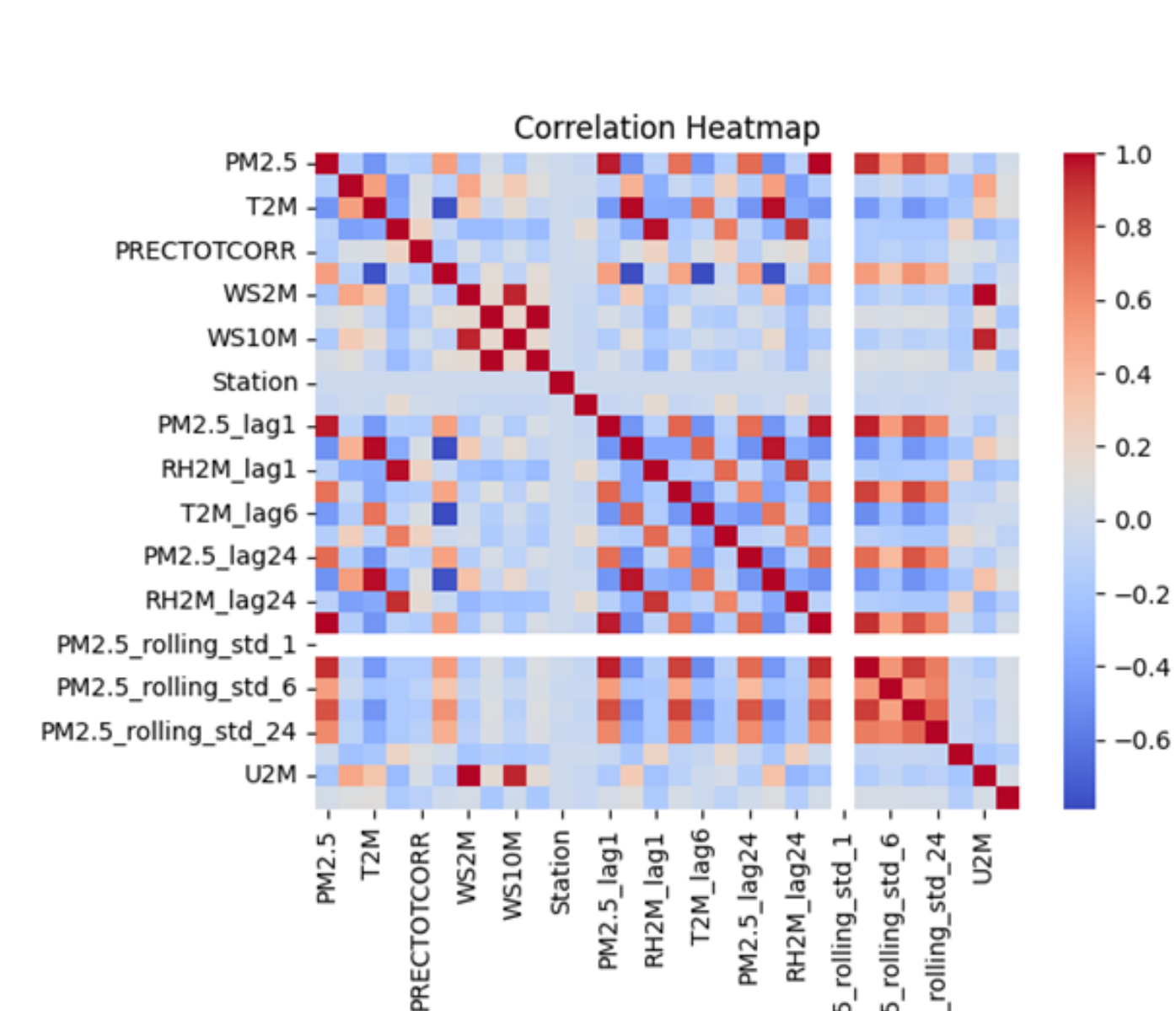


Figure 3: Correlation heatmap showing relationships between PM2.5 and meteorological parameters.

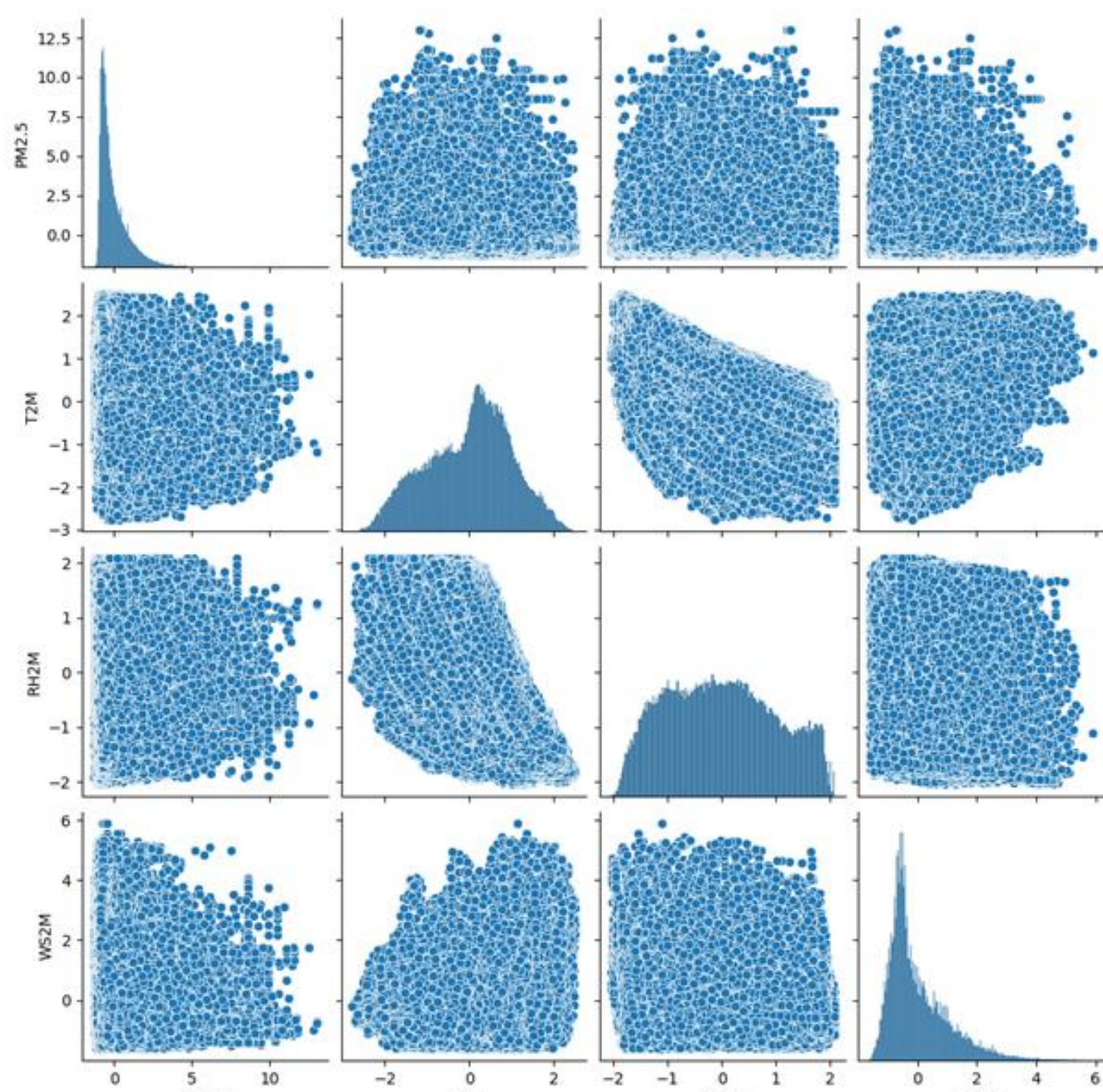


Figure 4: Scatter plots showing relationships between PM2.5, T2M, RH2M, and WS2M.

Modelling Approach

Deep Learning (DL): A subset of ML that uses multi-layer neural networks to learn complex patterns directly from raw data.

Model 1: PM2.5 Prediction Using Enhanced CNN + LSTM with Attention Mechanism

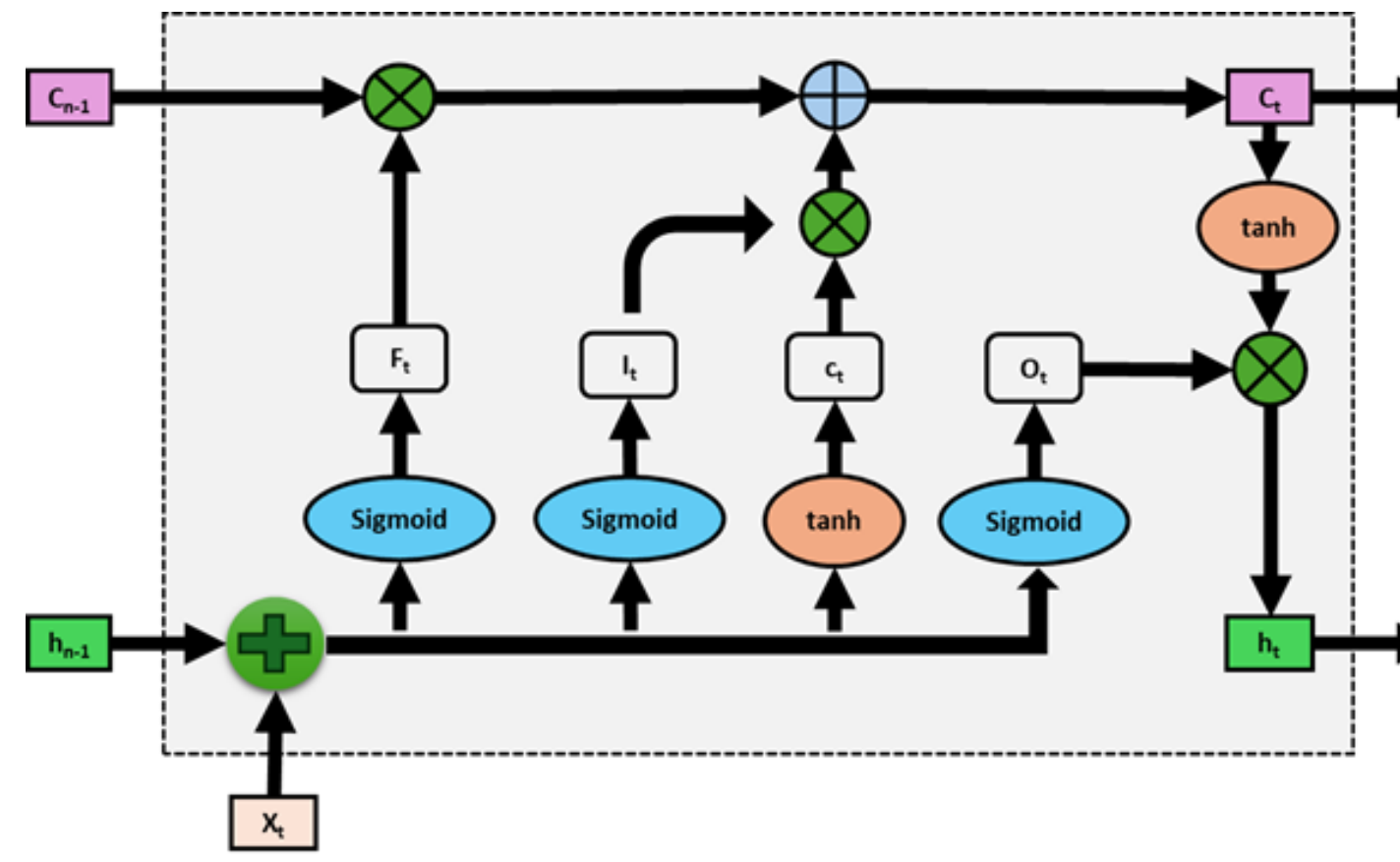


Figure 5: The architecture of the LSTM network.

Model 2: PM2.5 Prediction Using GRU and GNN

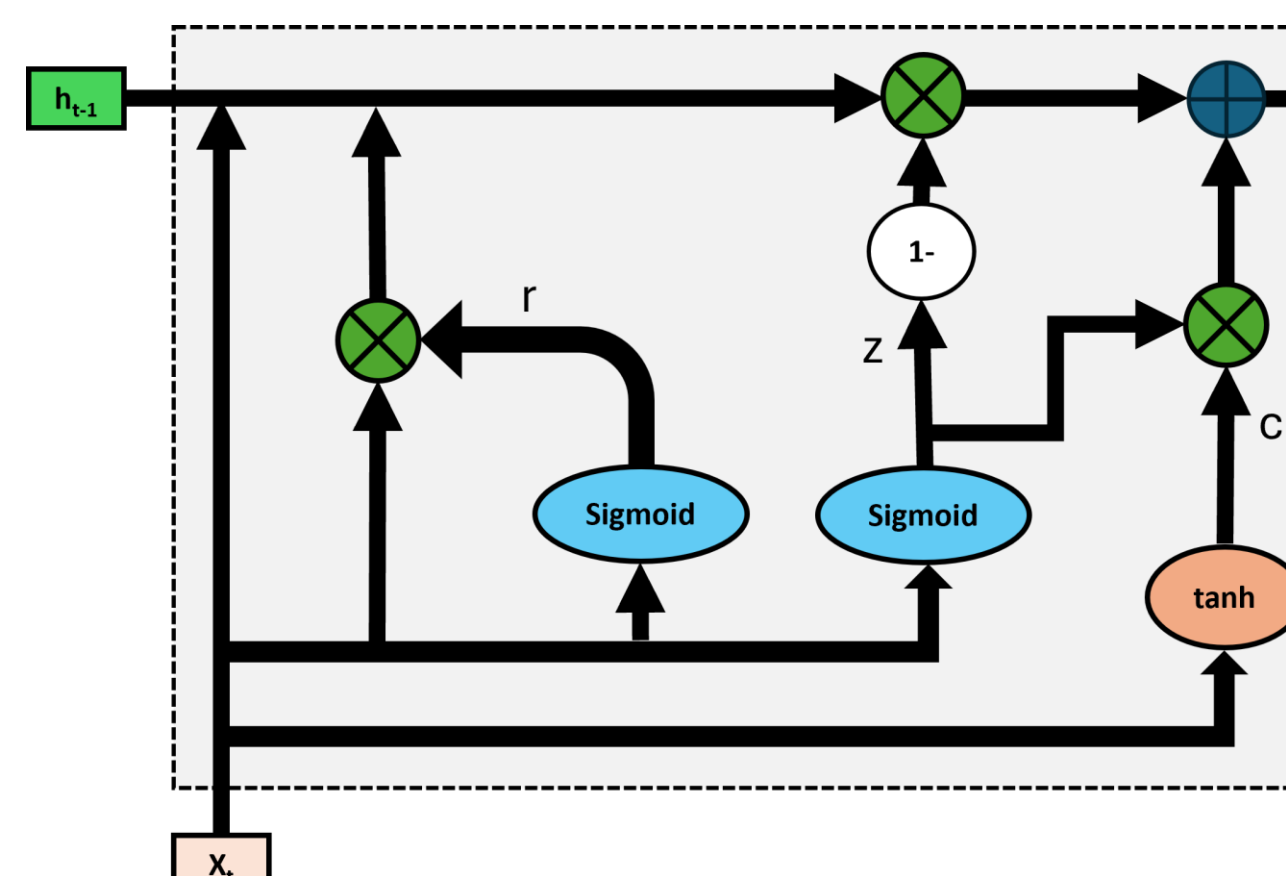


Figure 6: The architecture of the GRU network.

- CNNs: Specialized DL models that extract spatial features from grid-like data using convolutional and pooling layers.
- LSTMs: Recurrent networks with gating mechanisms to capture long-term dependencies in sequential data.

- GNNs: Extend DL to graph-structured data, learning from nodes, edges, and their relationships via message passing.
- GRUs: A simplified LSTM variant with fewer parameters, using update and reset gates for sequence modeling.

Results and Discussions

- Initial testing showed that both CNN-LSTM and GNN-GRU models needed tuning, with CNN-LSTM having an RMSE of 81.11 $\mu\text{g}/\text{m}^3$ and GNN-GRU an MAE of 47.38 $\mu\text{g}/\text{m}^3$ (after denormalization).
- After hyperparameter tuning, CNN-LSTM improved significantly to an RMSE of 26.67 $\mu\text{g}/\text{m}^3$, while GNN-GRU's MAE dropped to 37.36 $\mu\text{g}/\text{m}^3$.
- CNN-LSTM emerged as the better model, offering more accurate PM2.5 predictions than GNN-GRU.

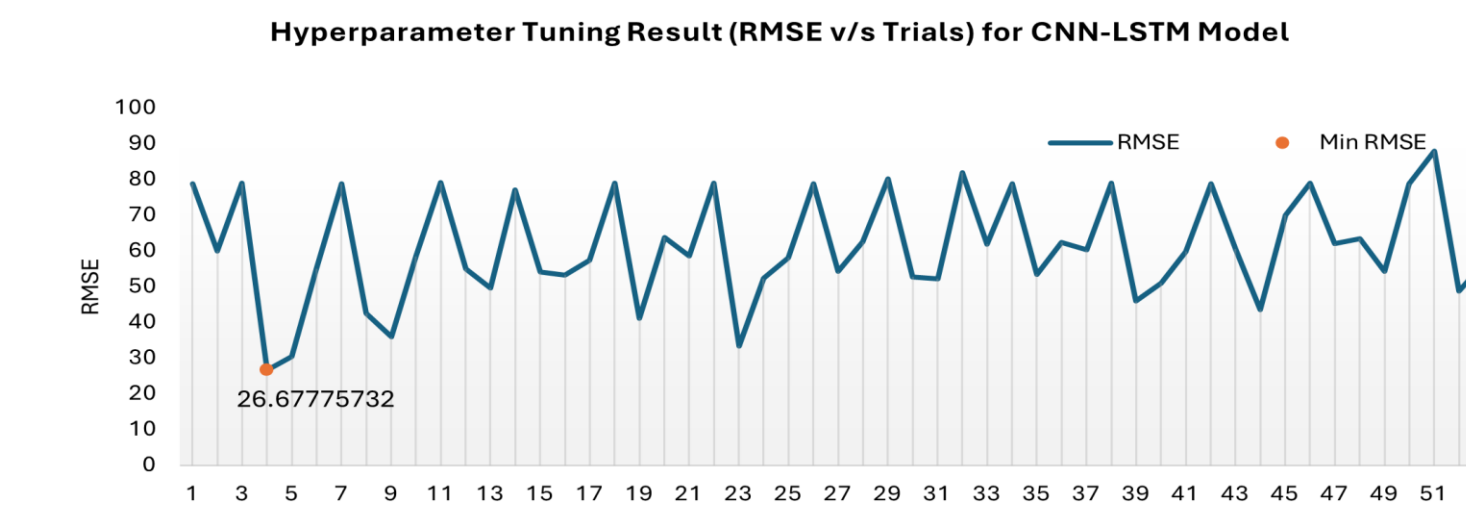


Figure 7: HPT results showing Error v/s Trials for CNN-LSTM Model.

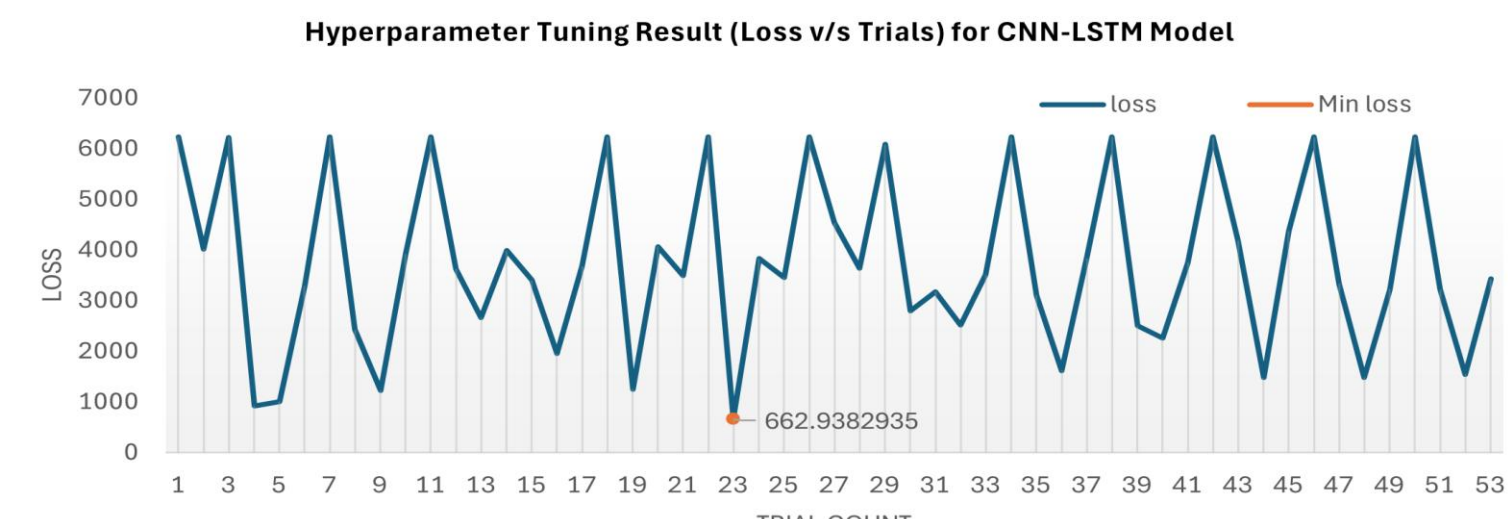


Figure 8: HPT Results showing Loss v/s Trials for CNN-LSTM Model.

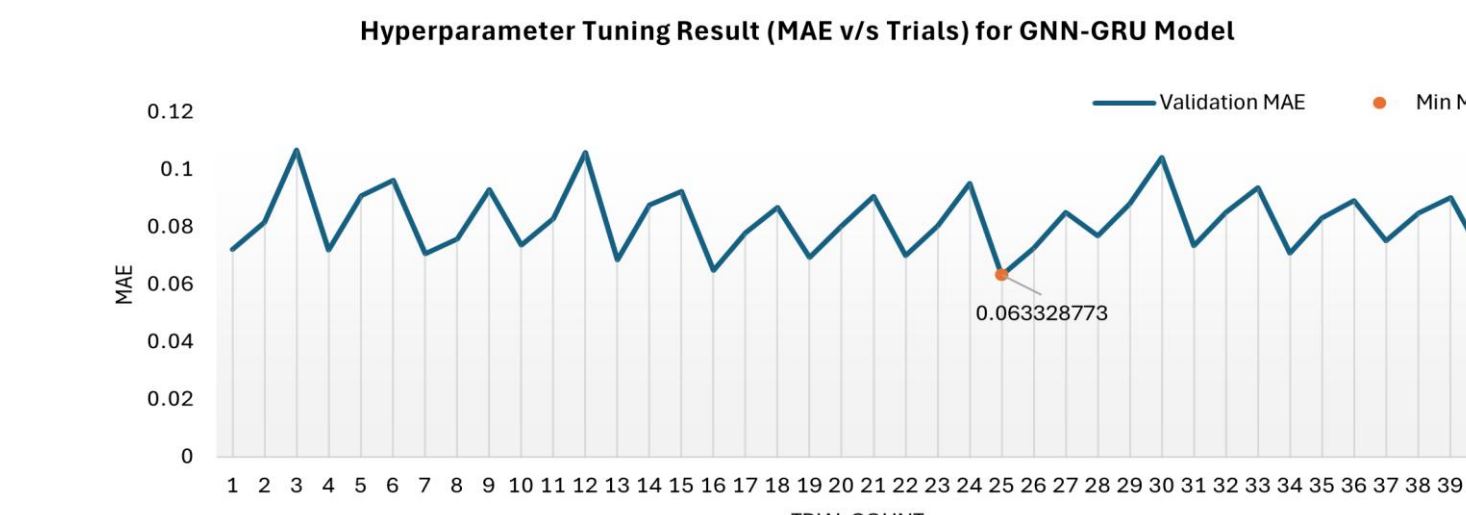


Figure 9: HPT results showing Error v/s Trials for GNN-GRU model.

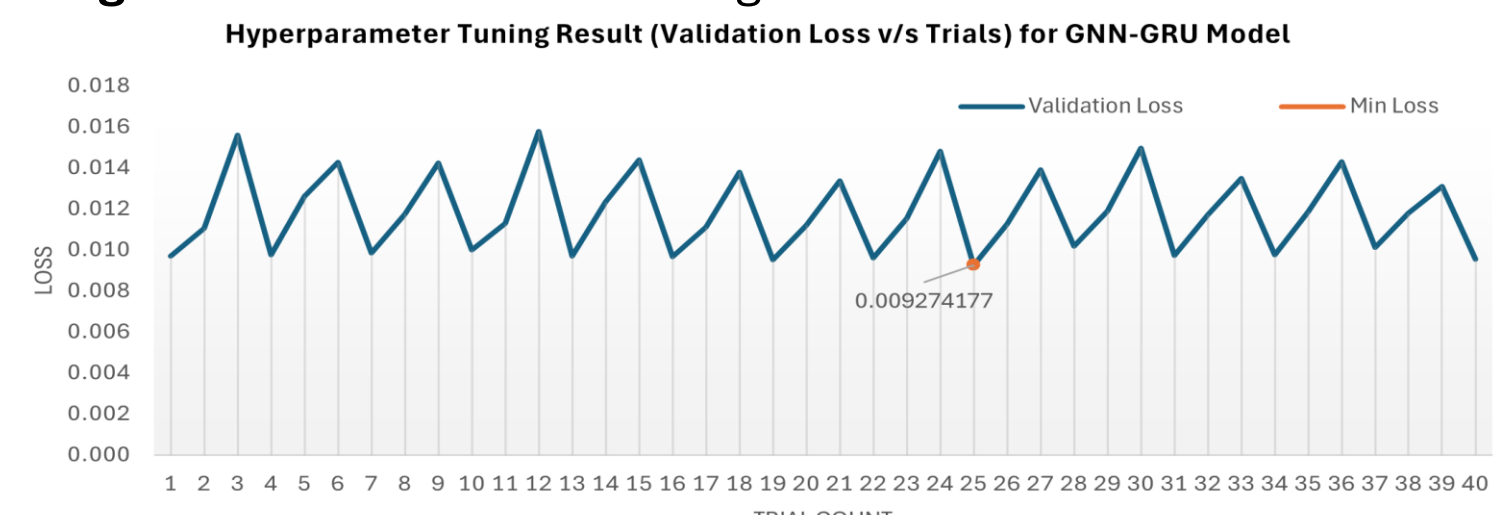


Figure 10: HPT results showing Loss v/s Trials for GNN-GRU model.

Detailed Analysis Of CNN-LSTM Model

Initial testing highlighted CNN-LSTM's strong performance, prompting a detailed evaluation. Scatter plots and time-series comparisons showed high alignment between actual and predicted PM2.5 values, with $R^2 = 0.78$ for 24-hour-ahead ($y[-1]$) and $R^2 = 0.80$ for 48-hour-ahead ($y[-2]$) forecasts. The model tracked trends well, especially at 48-hour lag, though minor deviations suggest room for improvement under varying conditions.

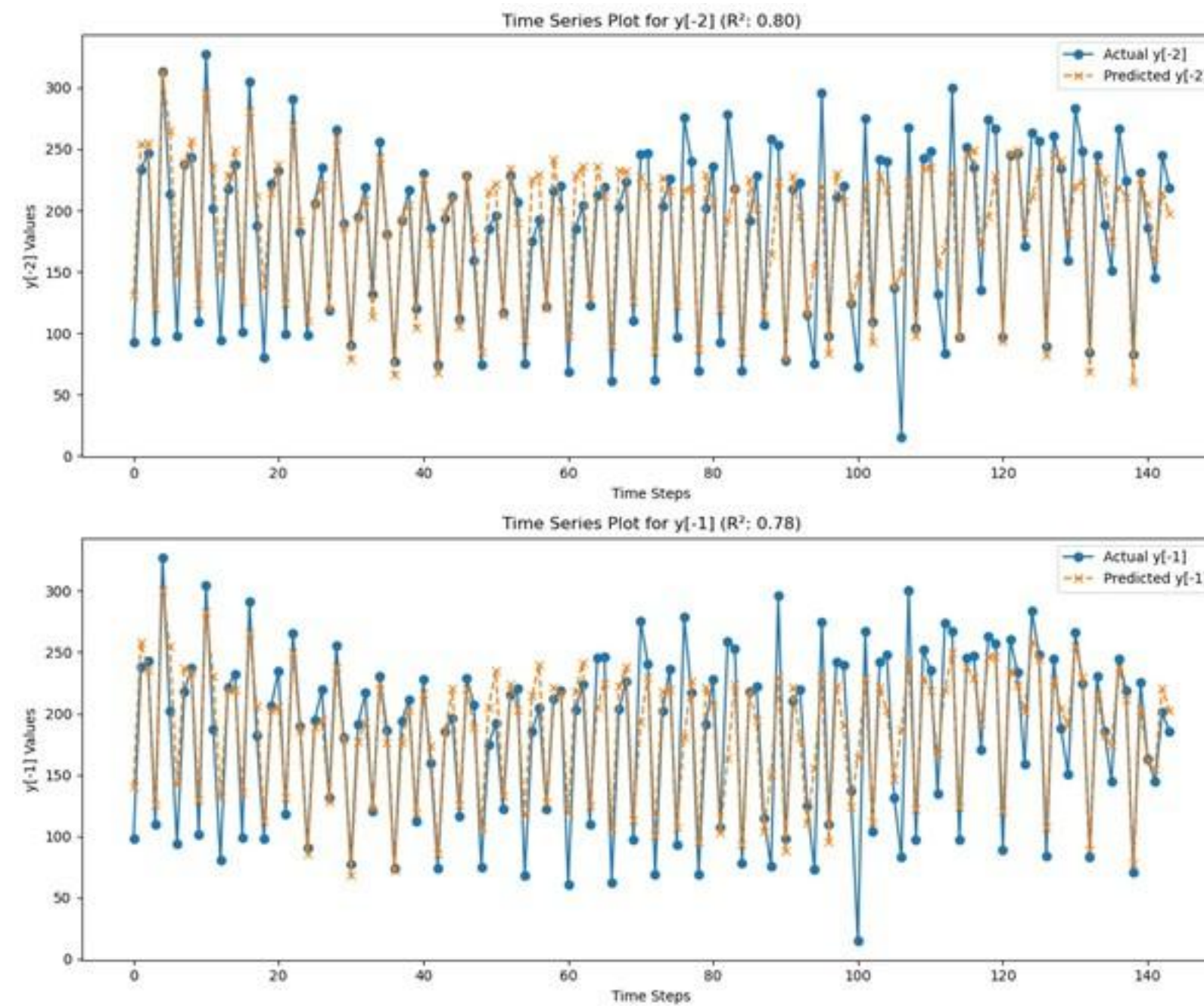


Figure 11: Time series plot for $y[-2]$ and $y[-1]$.

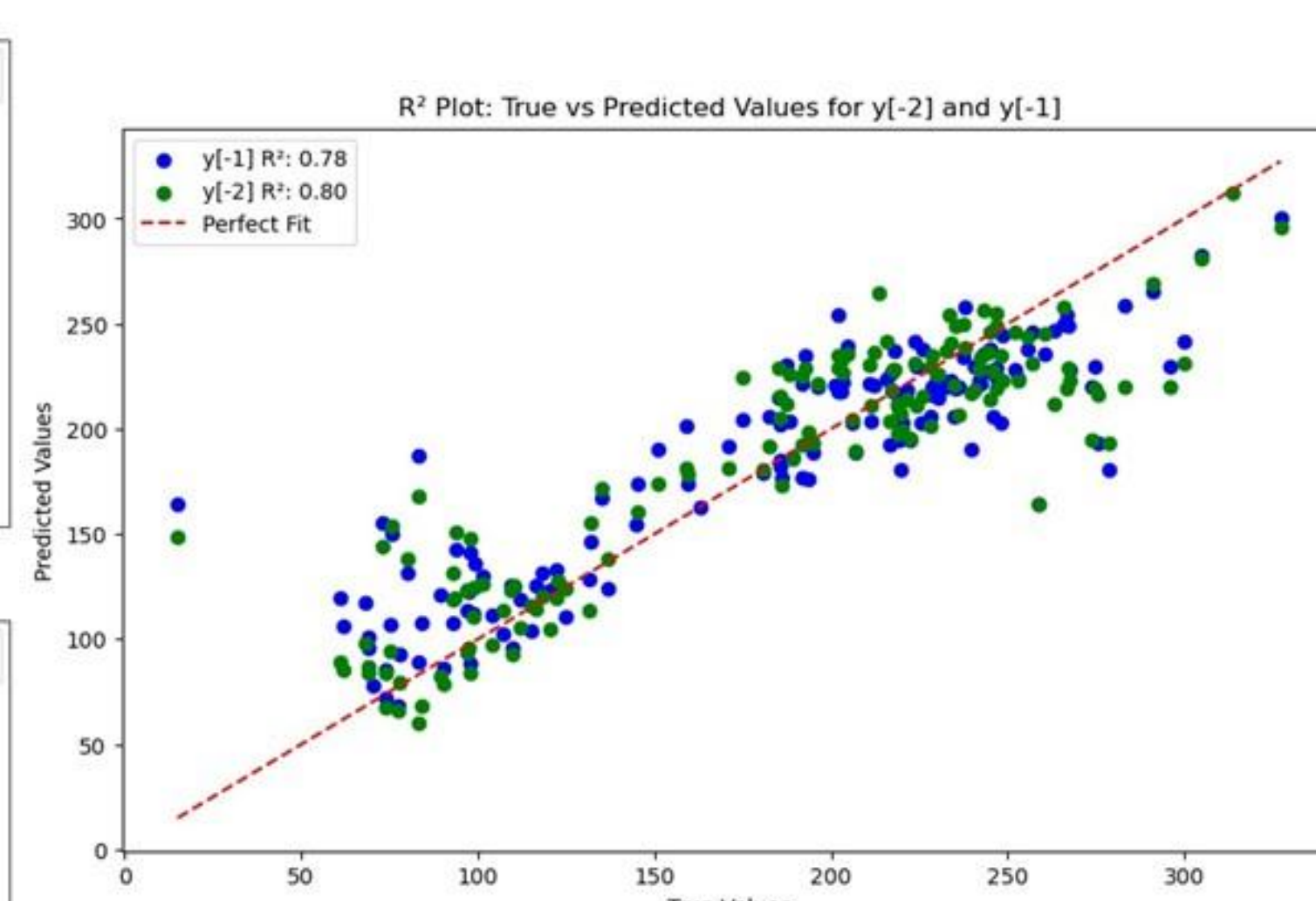


Figure 12: R-square plot between True and Predicted for $y[-2]$ and $y[-1]$.

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Acknowledgement

The authors gratefully acknowledge their parent research institute for providing essential support and resources throughout the course of this study. Special thanks are extended to the **Central Pollution Control Board (CPCB)** for providing access to high-quality air pollution data, and to **NASA's Data Access Viewer** for facilitating access to crucial meteorological datasets. Their contributions were instrumental in enabling the successful completion of this research.